

Course Title	Digital Signal Processing	Course Code	EE3031
Department	Department of Electrical Engineering (DEE)	Campus	Lahore
Knowledge Profile	Engineering Design (WK5)	Credit Hrs.	3+1
Knowledge Area	Telecommunication (KA12)	Grading Scheme	Relative
HEC Knowledge Area	Depth Electives	Applicable From	Spring 2023
SDG	4 Quality Education		
Pre-requisite(s)	EE/EL2008 Signals and Systems		

Course Objective	To introduce the fundamental techniques of digital signal processing and to develop ideas about the various aspects of digital signals and systems.
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No.	Assigned Program Learning Outcome (PLO)
1	An ability to apply knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.
3	An ability to design solutions for complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.

I = Introduction, R = Reinforcement, E = Evaluation, A = Assignment, Q = Quiz, M = Midterm, F=Final, L = Lab, P = Project, W = Written Report.

No.	Course Learning Outcome (CLO) Statements	Assessment Tools	Taxonomy Levels	PLO
1	Apply mathematical skills to analyze discrete time signals and systems.	M1,Q1	C3	1
2	Apply mathematical skills to perform discrete time signal operations.	A1,M1	C3	1
3	Apply discrete Fourier transform and its computational algorithm to analyze signals and systems	A2,M2,Q2	C3	3
4	Apply z-transform to analyze discrete time systems	Q3,M2,F	C3	3
5	Design filters using various techniques	A3,F	C5	3

Text Book(s)	Title	Digital Signal Processing Principles, Algorithms and Applications
	Author	John G. Proakis, Dimitris G. Manolakis
	Publisher	Prentice Hall, 4th edition
Ref. Book(s)	Title	Discrete-Time Signal Processing
	Author	Allen V. Oppenheim and Ronald W. Schaffer, Prentice Hall
	Publisher	

Week	Course Contents/Topics	Chapter*	CLO*
1	Discrete-time signal basics, discrete-time system basics and convolution of discrete-time signals	2.1-2.3	1,2
2	Autocorrelation and its application in radar signal detection	2.6	2
3	Introduction to Fourier analysis, frequency response and filtering, the Discrete-Time Fourier Transform (DTFT)	4	3
4	Properties of Discrete-Time Fourier Transform (DTFT), relationship between Discrete-Time Fourier Transform and Continuous-Time Fourier Transform (CTFT)	4	3
5	Review of D/A conversion, introduction to digital up sampling and down sampling, decimation and interpolation, multi-rate signal processing	6.1, 11.1, 11.4	2
6	Discrete Fourier Transform (DFT), periodic and circular convolutions	7	3
7	Frequency domain sampling, DFT based processing of a finite-length sine wave and time domain aliasing of multi-pole causal signals	7	3
8	Filtering using the DFT, overlap-add and overlap-save	7	3
9	Fast Fourier Transform (FFT) -in-time Radix 2 FFT, decimation-in-frequency Radix 2 FFT	8	3
10,11	Z-transform basics, the inverse Z-transform and stability analysis of discrete-time systems	3	4
12	graphical frequency response of filters	10	5
13	Filter design techniques	Lecture notes	5
14	Filter design	Lecture notes	5

Assessment Tools	Weightage
Quizzes, Assignments	20.0%
Midterms (I+II)	30%
Final Exam	50.0%